



# Cryptographic Protocol Verification and Analysis

University of Texas at Dallas - Team 1



## Team Members

### Student Investigators

Manasa Kandala

Pragathi Vemulapalli

Rahul Muthyam

Saloni Shivdasani

Shreyans Patel

### Technical Director

Dr. Ed Ziegler

### External Support

Enis Golaszewski

### Faculty Supervisor

Dr. Brian Ricks



# Agenda

- Project Motivation
- Project Overview
- Goals
- Current Progress
- Timeline
- Next Steps
- Q&A



## Project Motivation

- Standardization of cryptographic protocols lacks formal verification
- Protocol functionality is ensured, security is unreliable
- The community that works in this domain including us believe that carrying out such formal analysis using softwares like CPSA must become a standard practice for future protocol design



## Project Overview

In this project, we aim to:

1. Study Grant Authorization and Negotiation Protocol (GNAP)
2. Learn Cryptographic Protocol Shapes Analyzer (CPSA)
3. Use CPSA to analyze GNAP
4. Find potential solutions to any vulnerabilities (if found)



# Goals

- Analyze GNAP sequences that involve authentication and model them in CPSA
  - User-Code Interaction
  - Asynchronous Authorization
  - Cross-User Authentication
- Study the results and devise solutions if vulnerabilities are found



## Grant Negotiation and Authorization Protocol (GNAP)

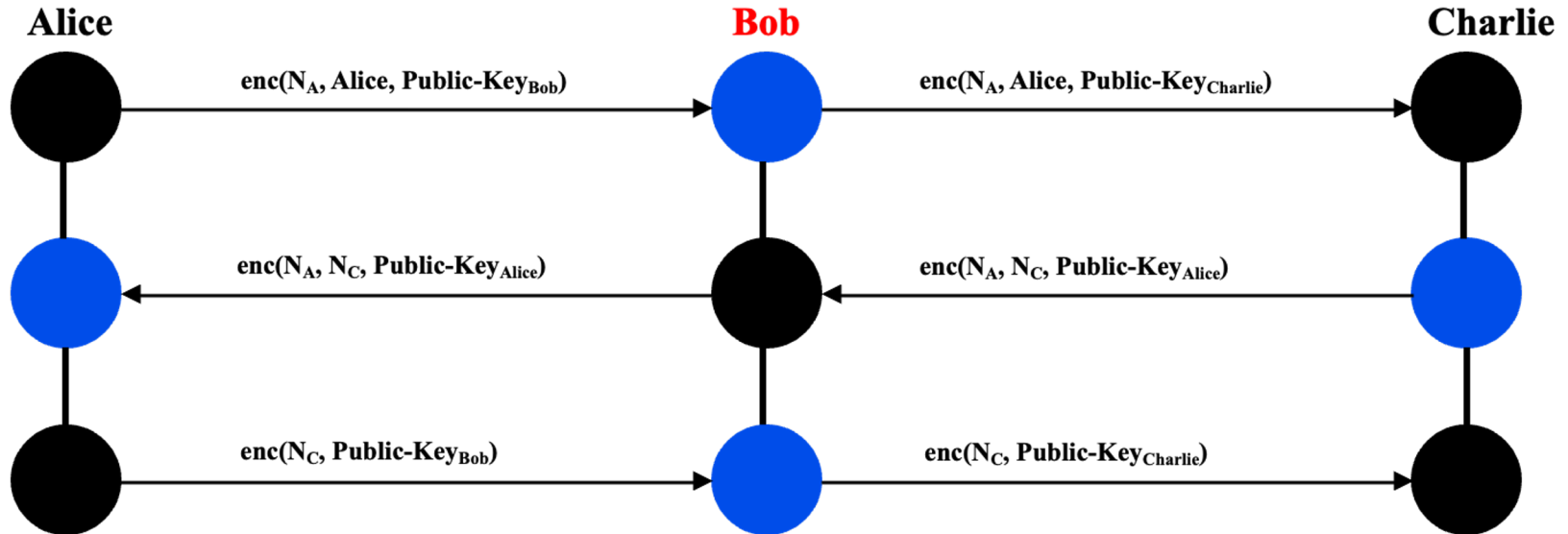
- GNAP is the next version of the OAuth2.x family of protocols
- It boasts several advantages over OAuth2
- Being relatively new and not yet standardized, GNAP is an excellent choice for formal analysis



## Cryptographic Protocol Shapes Analyzer (CPSA)

- CPSA is a software tool designed to assist in the design and analysis of cryptographic protocols
- It assumes the attacker can add/modify/flush any message that flows through the network i.e. the whole network is assumed to be compromised

## CPSA (Continued)

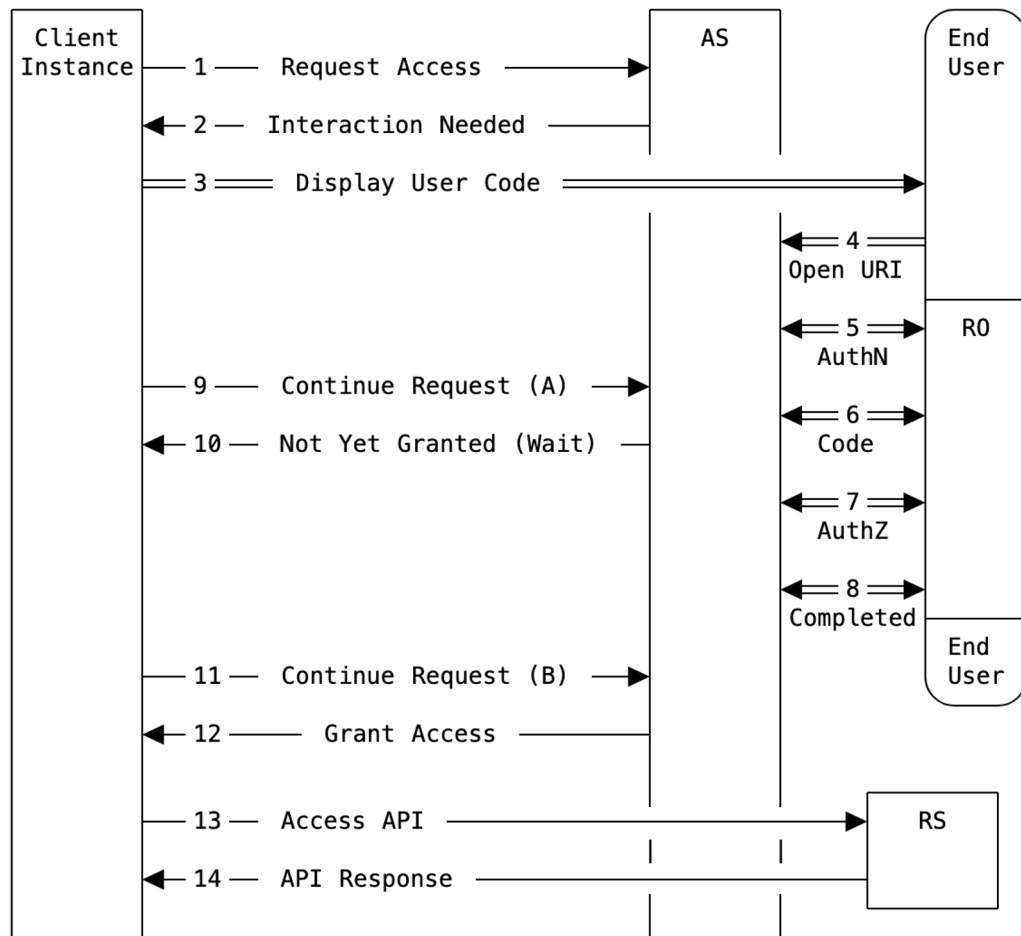


# GNAP Sequence

## User-Code Interaction

### GNAP Assumptions

- Communication channels are secure.
- No party is compromised.
- Client request are signed, server responses are not.





## Next Steps

Set-Targets:

- Complete the model with TLS macros, LTK and Channels feature.
- Analyze the model under various scenarios
- Study the generated graphs to find vulnerabilities and find solutions

Additional Targets (As per time constraints):

- Model additional GNAP sequences

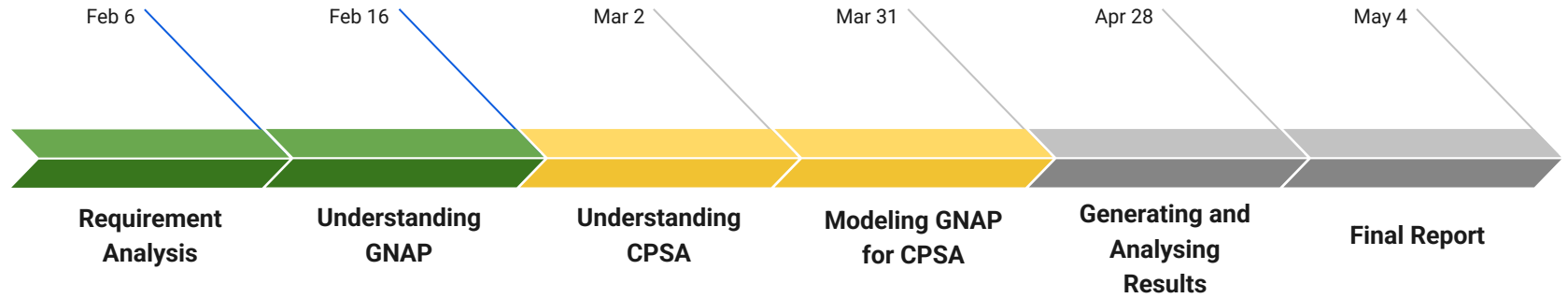


## Challenges

- Minor setback due to CPSA training delays
- Learning curve due to lack of experience with formal analysis



# Timeline



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Questions?